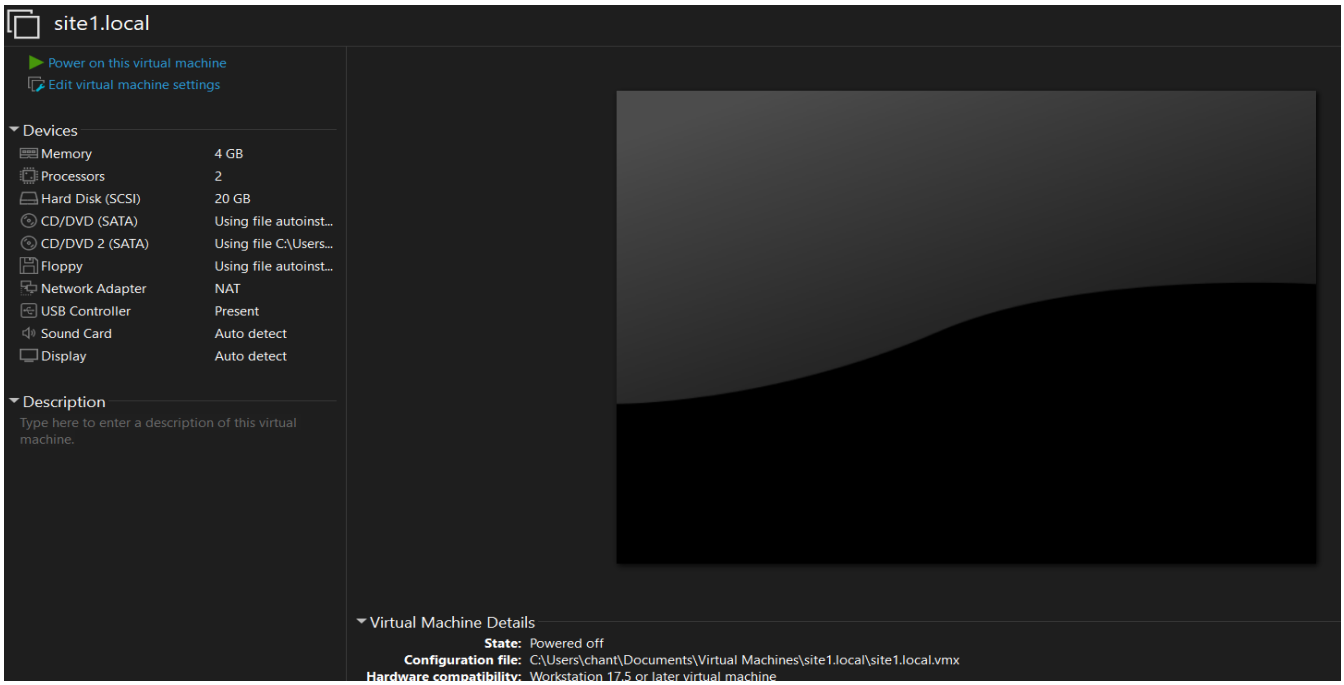
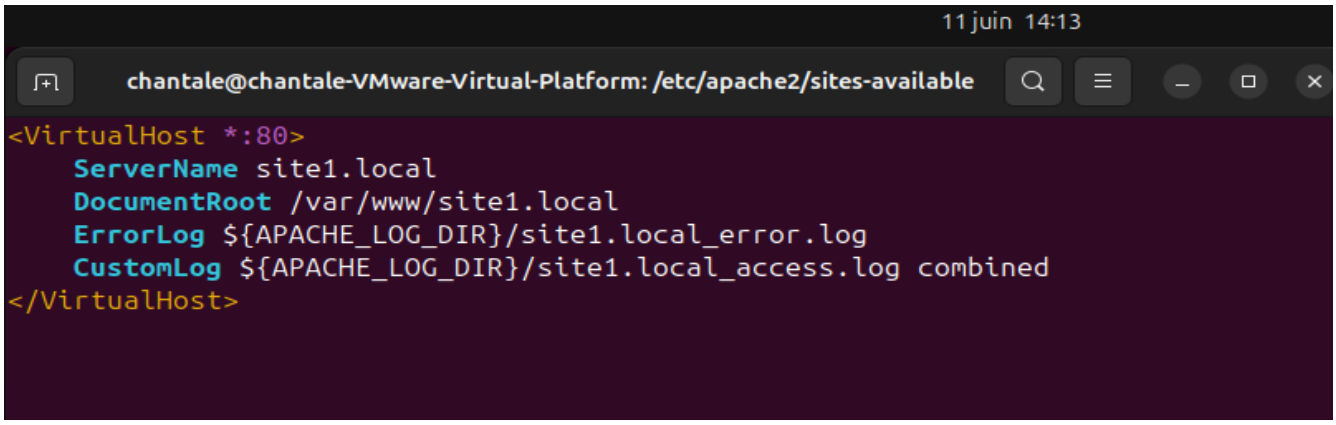
INTRODUCTION

Dans le cadre de mon projet au sein de l'entreprise PHOENIX, j'ai pour mission de mettre en place un hébergement multi-sites pour un client. L'objectif est de configurer deux sites PHP distincts avec chacun leur propre VirtualHost, leur pool PHP-FPM, un répertoire isolé et une base de données dédiée. Le projet comprend également la mise en place de restrictions d'accès, l'ajout de phpMyAdmin et l'automatisation des sauvegardes.

Création de la machine virtuelle avec l'iso Ubuntu



1.Un VirtualHost propre pour chaque site1.local



Un VirtualHost propre pour chaque site2.local

```
11 juin 16:11
chantale@chantale-VMware-Virtual-Platform: /etc/apache2/sites-available
<VirtualHost *80>
    ServerName site2.local
    DocumentRoot /var/www/site2.local
    ErrorLog ${APACHE_LOG_DIR}/site2.local_error.log
    CustomLog ${APACHE_LOG_DIR}/site2.local_access.log combined
</VirtualHost>
~
~
~
```

```
chantale@chantale-VMware-Virtual-Platform: /etc/apache2/sites-available
apache2/site1.local.conf
at: /etc/apache2/site1.local.conf: Aucun fichier ou dossier de ce nom
hantale@chantale-VMware-Virtual-Platform:/etc/apache2/sites-available$ cat site
.local.conf
VirtualHost *:80>
   ServerName site1.local
   DocumentRoot /var/www/site1.local
   ErrorLog ${APACHE_LOG_DIR}/site1_error.log
   CustomLog ${APACHE_LOG_DIR}/site1_access.log combined
/VirtualHost>
hantale@chantale-VMware-Virtual-Platform:/etc/apache2/sites-available$ sudo vim
ite2.local.conf
udo: vimsite2.local.conf : commande introuvable
hantale@chantale-VMware-Virtual-Platform:/etc/apache2/sites-available$ sudo vim
site2.local.conf
hantale@chantale-VMware-Virtual-Platform:/etc/apache2/sites-available$ cat site
.local.conf
VirtualHost *80>
   ServerName site2.local
   DocumentRoot /var/www/site2.local
   ErrorLog ${APACHE_LOG_DIR}/site2.local_error.log
   CustomLog ${APACHE_LOG_DIR}/site2.local_access.log combined
/VirtualHost>
hantale@chantale-VMware-Virtual-Platform:/etc/apache2/sites-available$
```

```
chantale@chantale-VMware-Virtual-Platform:/etc/apache2/sites-available$ sudo a2e
nsite site2.local.conf
Site site2.local already enabled
chantale@chantale-VMware-Virtual-Platform:/etc/apache2/sites-available$ sudo sys
temctl reload apache2
chantale@chantale-VMware-Virtual-Platform:/etc/apache2/sites-available$ sudo a2e
nsite site1.local.conf
Site site1.local already enabled
chantale@chantale-VMware-Virtual-Platform:/etc/apache2/sites-available$ sudo a2e
nsite site1.local.conf
Site site1.local already enabled
chantale@chantale-VMware-Virtual-Platform:/etc/apache2/sites-available$ sudo sys
temctl reload apache2
chantale@chantale-VMware-Virtual-Platform:/etc/apache2/sites-available$
```

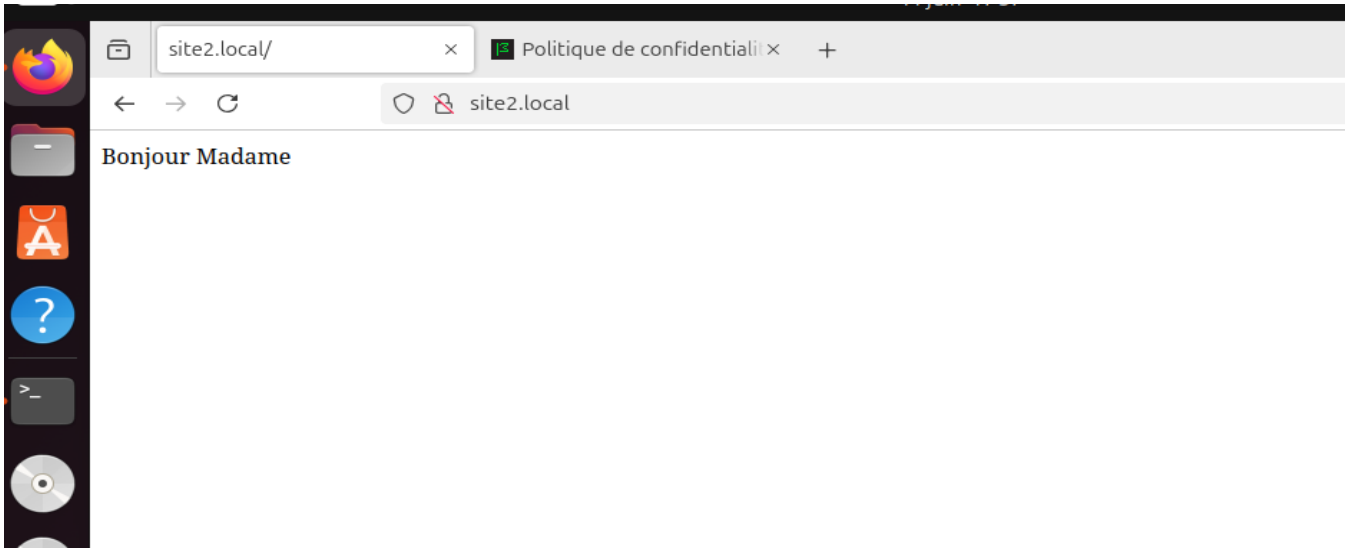
2- un dossier isolé pour chaque site

```
11 juin 16:54
chantale@chantale-VMware-Virtual-Platform: /var/www/site2.local
chantale@chantale-VMware-Virtual-Platform:/etc/apache2$ cd ..
chantale@chantale-VMware-Virtual-Platform:/etc$ cd ..
chantale@chantale-VMware-Virtual-Platform:/$ cd ..
chantale@chantale-VMware-Virtual-Platform:/$ cd /var/www/
chantale@chantale-VMware-Virtual-Platform:/var/www$ sudo mkdir site2.local
chantale@chantale-VMware-Virtual-Platform:/var/www$ ls
html  site2.local
chantale@chantale-VMware-Virtual-Platform:/var/www$ sudo mkdir site1.local
chantale@chantale-VMware-Virtual-Platform:/var/www$ ls
html  site1.local  site2.local
chantale@chantale-VMware-Virtual-Platform:/var/www$ cd site1.local/
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$ touch index.html
touch: impossible de faire un touch 'index.html': Permission non accordée
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$  sudo touch inde
x.html
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$ ls
index.html
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$ cd ..
chantale@chantale-VMware-Virtual-Platform:/var/www$ cd site2.local/
chantale@chantale-VMware-Virtual-Platform:/var/www/site2.local$ sudo touch index
.html
chantale@chantale-VMware-Virtual-Platform:/var/www/site2.local$ ls
index.html
chantale@chantale-VMware-Virtual-Platform:/var/www/site2.local$
```

je donne une adresse ip à chaque site dans mon fichier vhosts

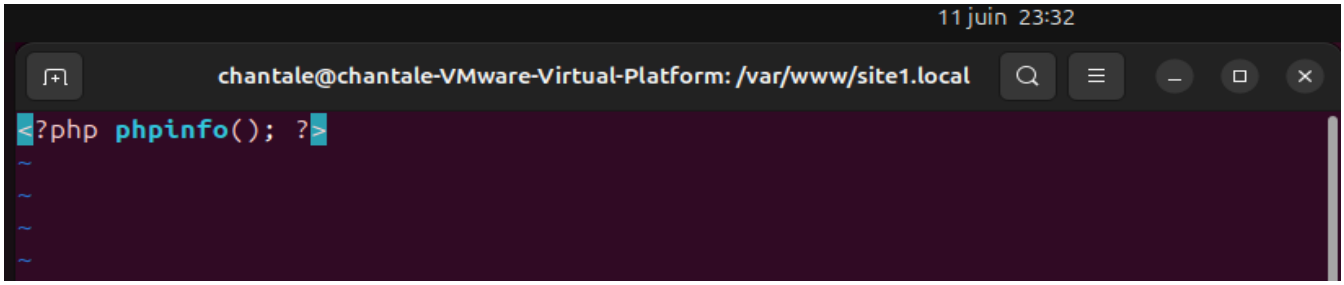
```
chantale@chantale-VMware-Virtual-Platform: ~
127.0.0.1 localhost
127.0.0.2 chantale-VMware-Virtual-Platform
127.0.0.3 site1.local
127.0.0.4 site2.local
# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
~
~
~
~
~
~
~
~
~
~
```

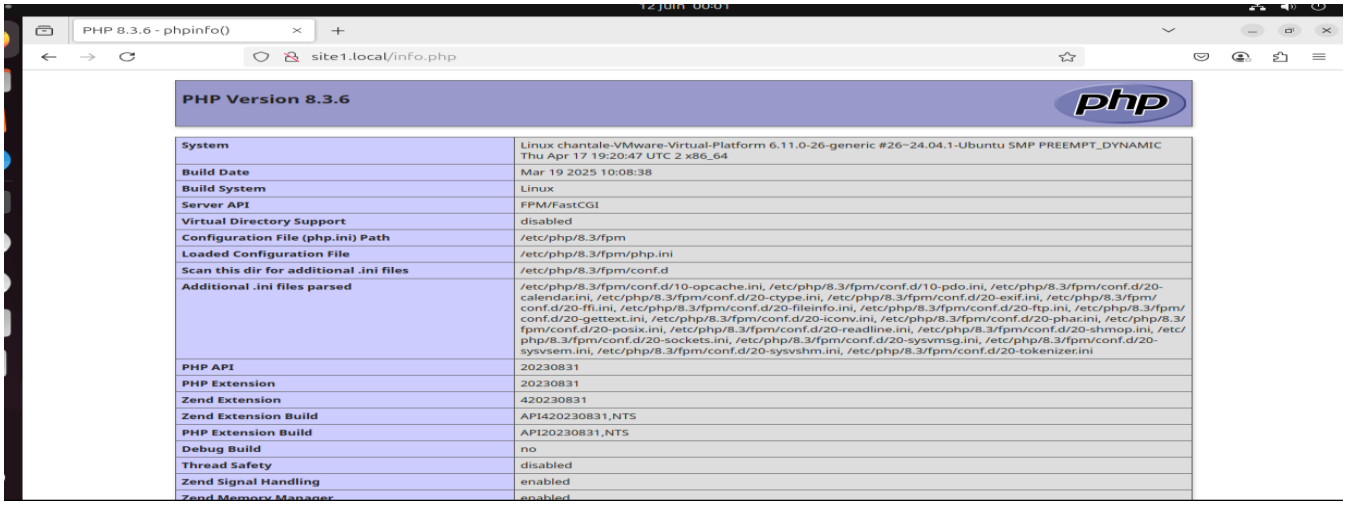
Je recherche mes sites sur mon navigateur

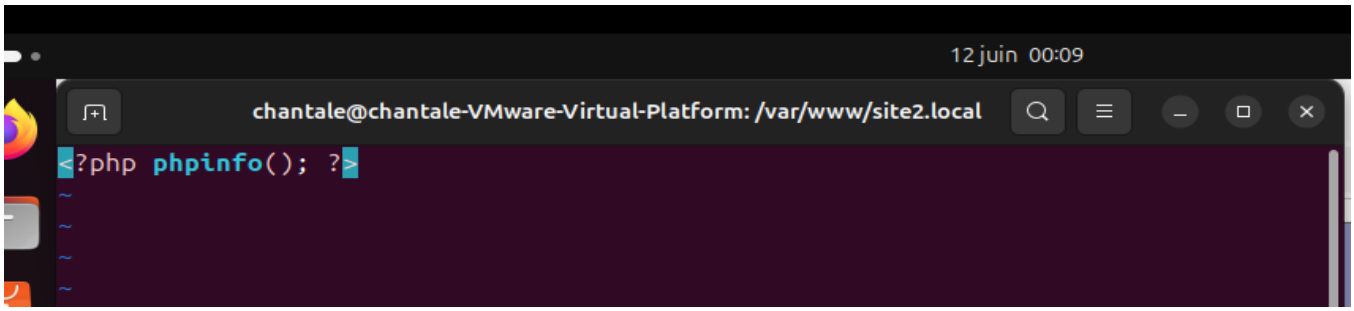


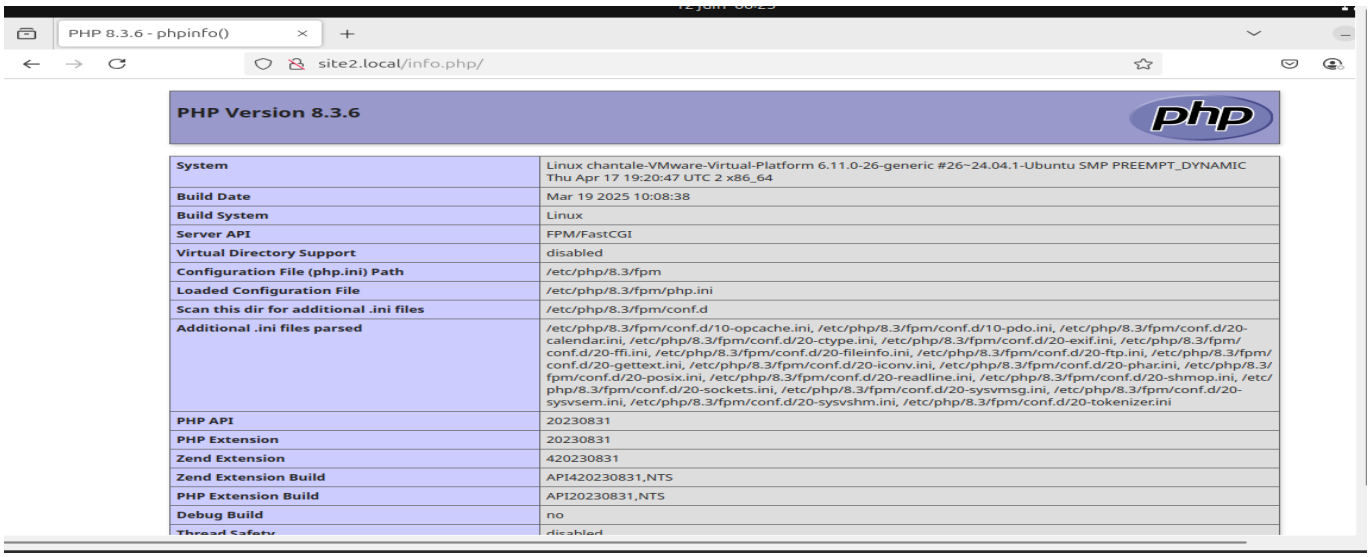
3-Une page qui affiche les informations sur la partie PHP

```
chantale@chantale-VMware-Virtual-Platform:~$ cd /var/www/site1.local/
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$ sudo vim info.ph
p
```



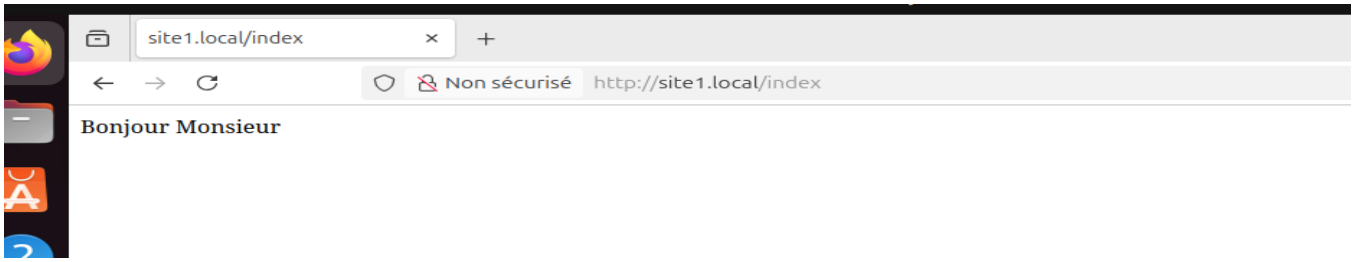






4-Une redirection de la page index vers index.html

```
chantale@chantale-VMware-Virtual-Platform: ~
<VirtualHost *:80>
    ServerName site1.local
    DocumentRoot /var/www/site1.local
    ErrorLog ${APACHE_LOG_DIR}/site1_error.log
    CustomLog ${APACHE_LOG_DIR}/site1_access.log combined
    RewriteEngine on
    RewriteRule index index.html [NC]
</VirtualHost>
~
~
```



5-Restriction de la page phpinfo pour les utilisateurs bob et alice

J'ai d'abord créer un fichier .htaccess

```
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$ sudo touch .hta
ccess
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$ ls -a
.  ..  .htaccess  index.html  info.php
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$ sudo vim htacces
s
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$
```

J'ai écrit à l'intérieur de mon fichier

```
12 juin 13:52
chantale@chantale-VMware-Virtual-Platform: /var/www/site1.local
AuthType Basic
Auth Name "Zone protégée"
AuthUserFile /etc/apache2/htpasswd
Require bob alice
~
~
~
```

J'ai creer les utilisateurs bob et alice

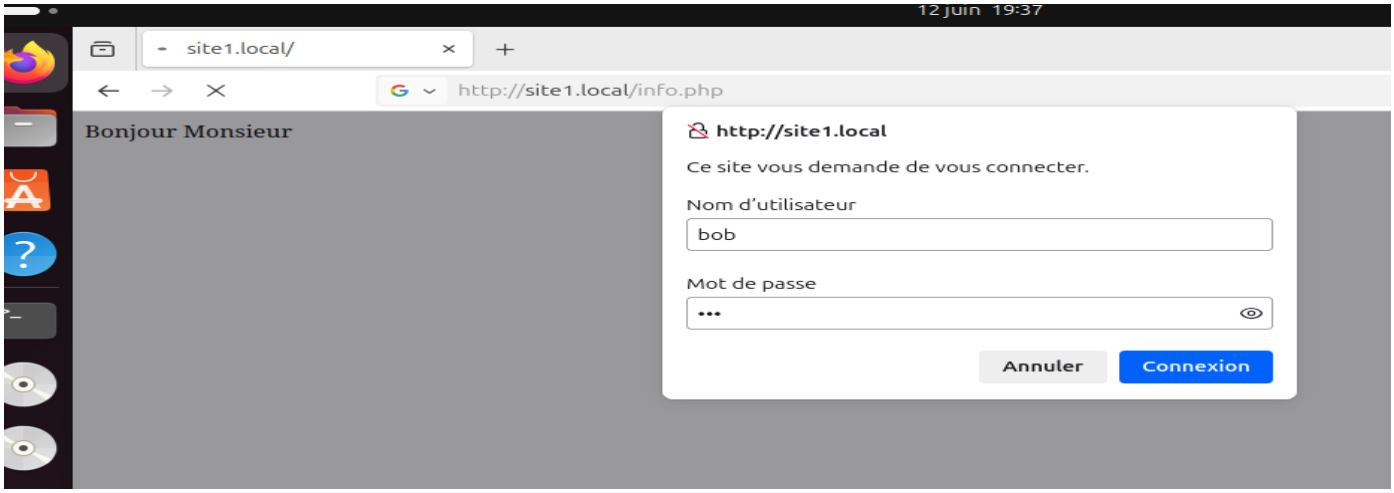
```
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$ sudo htpasswd -c /etc/apache2/.htpasswd bob
New password:
Re-type new password:
Adding password for user bob
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$
```

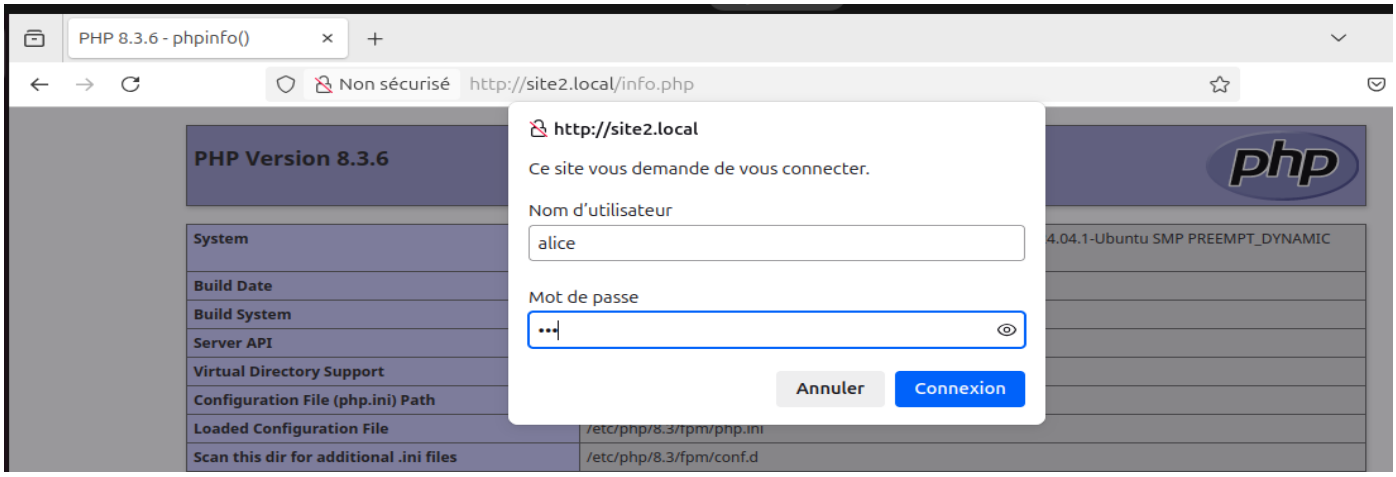
```
The SHA algorithm does not use a salt and is less secure than the MD5 algorithm.
chantale@chantale-VMware-Virtual-Platform:/var/www/site1.local$ sudo htpasswd  /etc/apache2/.htpasswd alice
New password:
Re-type new password:
Adding password for user alice
Afficher les applications
Mware-Virtual-Platform:/var/www/site1.local$
```

J'ai rajouter sur mes fichiers des virthualhost mes lignes de restriction

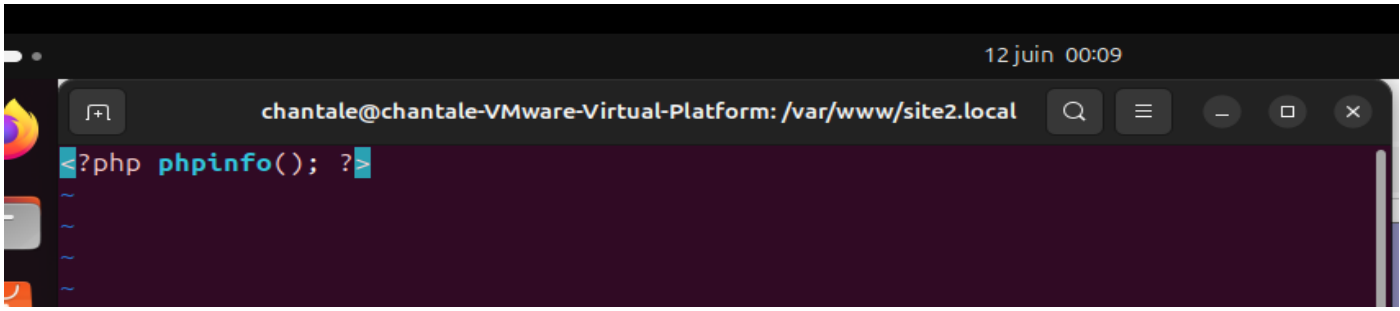
```
12 juin 17:51
chantale@chantale-VMware-Virtual-Platform: /var/www/site1.local
<VirtualHost *:80>
    ServerName site1.local
    DocumentRoot /var/www/site1.local
    ErrorLog ${APACHE_LOG_DIR}/site1_error.log
    CustomLog ${APACHE_LOG_DIR}/site1_access.log combined
    RewriteEngine on
    RewriteRule index index.html [NC]
<Files>
    AllowOverride All
    Require all granted
    AuthType Basic
    AuthName "Zone protégée"
    AuthUserFile /etc/apache2/ .htpasswd
    Require bob alice
</Files>


</VirtualHost>
```

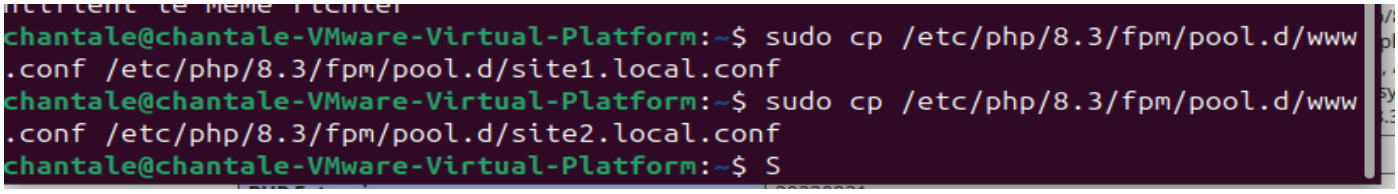
```
chantale@chantale-VMware-Virtual-Platform: /var/www/site2.local
<VirtualHost *:80>
    ServerName site2.local
    DocumentRoot /var/www/site2.local
    ErrorLog ${APACHE_LOG_DIR}/site2.local_error.log
    CustomLog ${APACHE_LOG_DIR}/site2.local_access.log combined

<Files>
    AuthType Basic
    AuthName "Zoneprotégée"
    AuthUserFile /etc/apache2/.htpasswd
    Require bob alice
    Require all granted
</Files>

</VirtualHost>
~
~
~
~
~
```





6- un pool PHP-FPM séparé pour chaque site





```
chantale@chantale-VMware-Virtual-Platform: ~
 Start a new pool named 'www'.
 the variable $pool can be used in any directive and will be replaced by the
 pool name ('www' here)
[site1]

 Per pool prefix
 It only applies on the following directives:
 - 'access.log'
 - 'slowlog'
 - 'listen' (unixsocket)
 - 'chroot'
 - 'chdir'
 - 'php_values'
 - 'php_admin_values'
 When not set, the global prefix (or /usr) applies instead.
 Note: This directive can also be relative to the global prefix.
 Default Value: none
prefix = /path/to/pools/$pool

 Unix user/group of the child processes. This can be used only if the master
 process running user is root. It is set after the child process is created.
 The user and group can be specified either by their name or by their numeric
 IDs.
-- INSERTION --                                          1,31          Haut
```

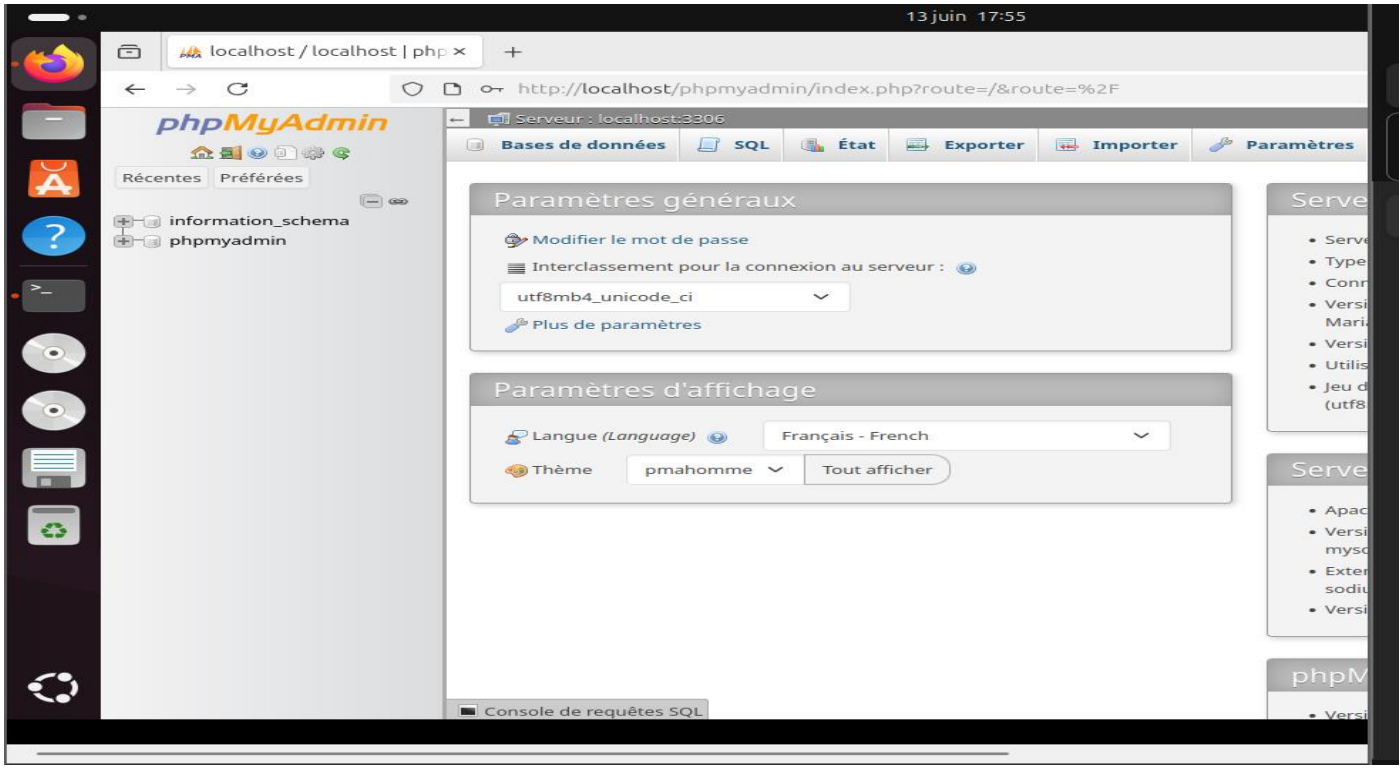
```
                          (IPv6 and IPv4-mapped) on a specific port;
   '/path/to/unix/socket' - to listen on a unix socket.
 Note: This value is mandatory.
listen = /run/php/php8.3-site1-fpm.sock

 Set listen(2) backlog.
 Default Value: 511 (-1 on Linux, FreeBSD and OpenBSD)
listen.backlog = 511

 Set permissions for unix socket, if one is used. In Linux, read/write
 permissions must be set in order to allow connections from a web server. Many
 BSD-derived systems allow connections regardless of permissions. The owner
 and group can be specified either by name or by their numeric IDs.
 Default Values: Owner is set to the master process running user. If the group
                 is not set, the owner's group is used. Mode is set to 0660.
listen.owner = www-data
listen.group = www-data
listen.mode = 0660

 When POSIX Access Control Lists are supported you can set them using
 these options, value is a comma separated list of user/group names.
 When set, listen.owner and listen.group are ignored
listen.acl_users =
-- INSERTION --                                          55,20          7%
```

```
chantale@chantale-VMware-Virtual-Platform: ~
; Start a new pool named 'www'.
; the variable $pool can be used in any directive and will be replaced by the
; pool name ('www' here)
[site2]

; Per pool prefix
; It only applies on the following directives:
; - 'access.log'
; - 'slowlog'
; - 'listen' (unixsocket)
; - 'chroot'
; - 'chdir'
; - 'php_values'
; - 'php_admin_values'
; When not set, the global prefix (or /usr) applies instead.
; Note: This directive can also be relative to the global prefix.
; Default Value: none
;prefix = /path/to/pools/$pool

; Unix user/group of the child processes. This can be used only if the master
; process running user is root. It is set after the child process is created.
; The user and group can be specified either by their name or by their numeric
; IDs.
-- INSERTION --                                          4,7           Haut
```

```
chantale@chantale-VMware-Virtual-Platform: ~
 The address on which to accept FastCGI requests.
 Valid syntaxes are:
   'ip.add.re.ss:port'     - to listen on a TCP socket to a specific IPv4 addres
 on
                             a specific port;
   '[ip:6:addr:ess]:port'  - to listen on a TCP socket to a specific IPv6 addres
 on
                             a specific port;
   'port'                  - to listen on a TCP socket to all addresses
                             (IPv6 and IPv4-mapped) on a specific port;
   '/path/to/unix/socket'  - to listen on a unix socket.
 Note: This value is mandatory.
isten = /run/php/php8.3-site2-fpm.sock

 Set listen(2) backlog.
 Default Value: 511 (-1 on Linux, FreeBSD and OpenBSD)
listen.backlog = 511

 Set permissions for unix socket, if one is used. In Linux, read/write
 permissions must be set in order to allow connections from a web server. Many
 BSD-derived systems allow connections regardless of permissions. The owner
 and group can be specified either by name or by their numeric IDs.
 Default Values: Owner is set to the master process running user. If the group
- INSERTION --                                          41,31          6%
```

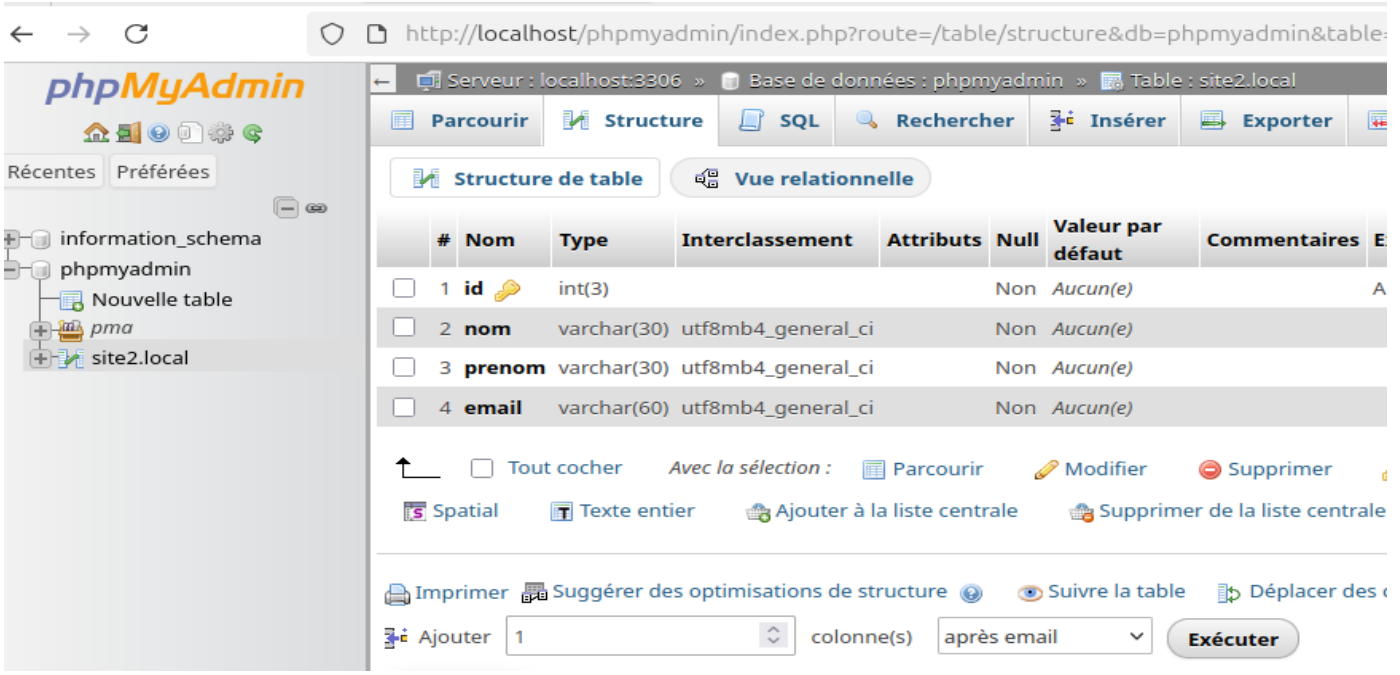
6-Interface de phpMyAdmin pour gérer les bases et les utilisateurs

```
chantale@chantale-VMware-Virtual-Platform:/var/www/site2.local$ sudo apt install phpmyadm
[sudo] Mot de passe de chantale :
```



Création de la base de données :phpmyadmin

Création de la table site2.local



Site2.local avec une page phoenix.html

index.html | ville.html | ville.css | phoenix.html | phoenix.css | style.css | info

phoenix.html > html > body > section#services.services > div.container > ul > li

```html
<!DOCTYPE html>
<html lang="fr">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Entreprise PHOENIX</title>
  <link rel="stylesheet" href="phoenix.css" />
</head>
<body>
  <header>
    <div class="container">
      <h1>PHOENIX</h1>
      <nav>
        <ul>
          <li><a href="#about">À propos</a></li>
          <li><a href="#services">Services</a></li>
          <li><a href="#contact">Contact</a></li>
        </ul>
      </nav>
    </div>
  </header>

  <section class="hero">
    <div class="container">
      <h2>Renaissance de vos projets numériques</h2>
      <p>PHOENIX, votre partenaire en solutions web et digitales.</p>
    </div>
  </section>

  <section id="about" class="about">
    <div class="container">
      <h2>À propos de nous</h2>
      <p>Nous sommes une entreprise innovante spécialisée dans le développement we
    </div>
```

```html
phoenix.html > html > body > section#services.services > div.container > ul > li
<html lang="fr">
<body>

  <section id="about" class="about">
    <div class="container">
      <h2>À propos de nous</h2>
      <p>Nous sommes une entreprise innovante spécialisée dans le développement we
    </div>
  </section>

  <section id="services" class="services">
    <div class="container">
      <h2>Nos Services</h2>
      <ul>
        <li>Sécurité informatique</li>
        <li>Développement web</li>
        <li>Solutions cloud</li>
      </ul>
    </div>
  </section>

  <section id="contact" class="contact">
    <div class="container">
      <h2>Contactez-nous</h2>
      <p>Email : contact@phoenix.fr</p>
      <p>Téléphone : 01 23 45 67 89</p>
    </div>
  </section>

  <footer>
    <div class="container">
      <p>&copy; 2025 PHOENIX. Tous droits réservés.</p>
    </div>
  </footer>
</body>
Ln 43, Col 13   Spaces: 4   UTF-8   LF   {} HTML
```

La page phoenix.css

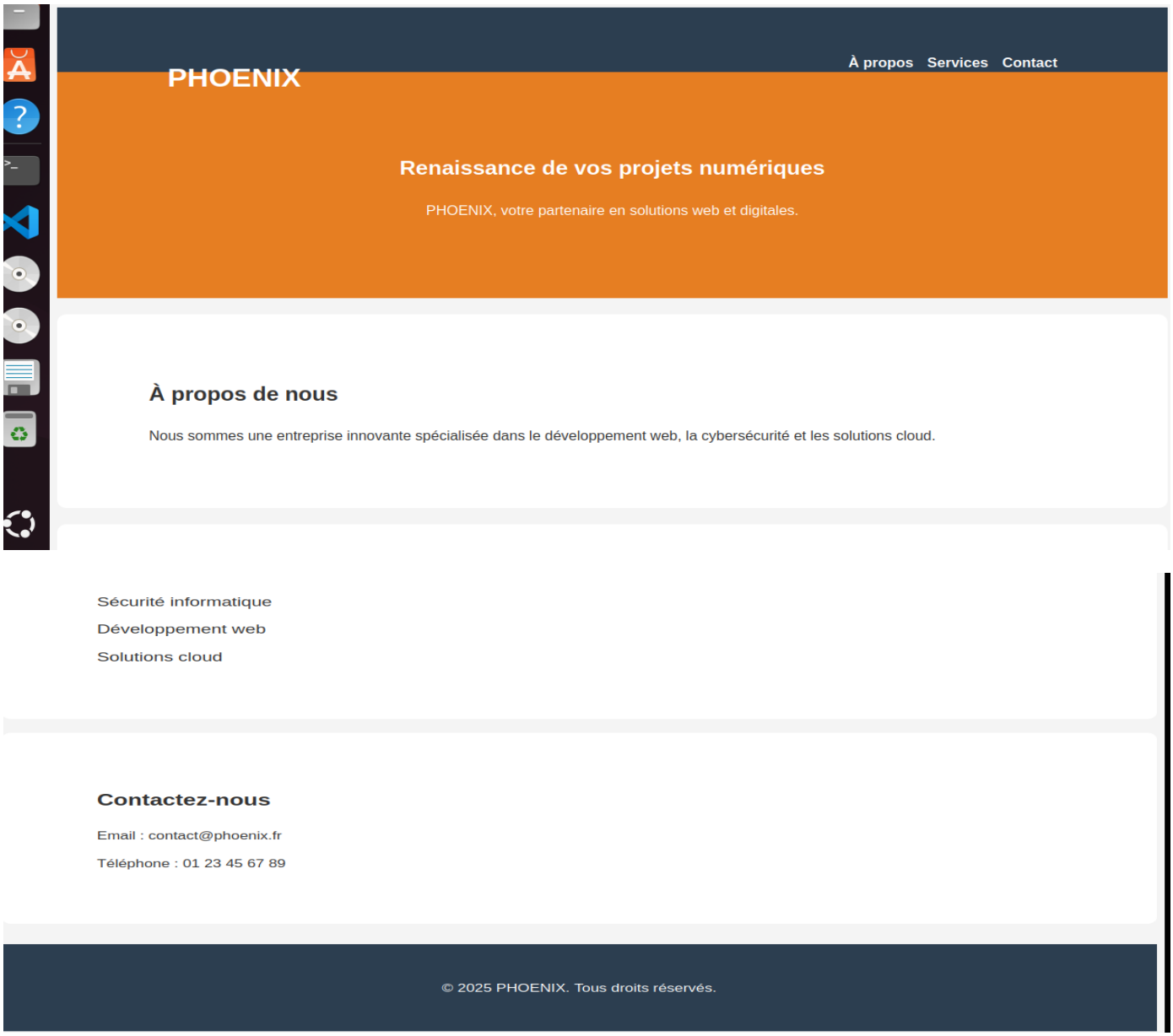
phoenix.css > ...

```css

body {
  font-family: Arial, sans-serif;
  line-height: 1.6;
  background-color: #f4f4f4;
  color: #333;
}

.container {
  width: 90%;
  max-width: 1000px;
  margin: auto;
  padding: 20px;
}

header {
  background: #2c3e50;
  color: white;
  padding: 20px 0;
}

header h1 {
  float: left;
  margin-left: 20px;
}

nav {
  float: right;
  margin-right: 20px;
}

nav ul {
  list-style: none;
  display: flex;
```

phoenix.css > ...

```css
nav a {
}

.hero {
  background: #e67e22;
  color: white;
  padding: 60px 20px;
  text-align: center;
}

.about, .services, .contact {
  background: white;
  margin-top: 20px;
  padding: 40px 20px;
  border-radius: 10px;
}

.services ul {
  list-style: none;
  padding-left: 0;
}

.services li {
  margin: 10px 0;
  font-size: 1.2em;
}

footer {
  background: #2c3e50;
  color: white;
  text-align: center;
  padding: 15px 0;
  margin-top: 30px;
}
```

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} CSS



Les fichiers logs distincts pour les 2 sites

Site1.local

```
hantale@chantale-VMware-Virtual-Platform:/var/log/apache2$ cat site1_access.log
27.0.0.1 - - [15/Jun/2025:11:48:24 +0200] "GET /index.html HTTP/1.1" 304 248 "-
 "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:139.0) Gecko/20100101 Firefox/139.
"
27.0.0.1 - - [15/Jun/2025:11:51:02 +0200] "GET /index.html HTTP/1.1" 200 300 "-
 "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:139.0) Gecko/20100101 Firefox/139.
"
27.0.0.1 - - [15/Jun/2025:19:17:06 +0200] "GET /index.html HTTP/1.1" 200 300 "-
 "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:139.0) Gecko/20100101 Firefox/139.
"
27.0.0.1 - - [15/Jun/2025:19:17:08 +0200] "GET /index.html HTTP/1.1" 304 248 "-
 "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:139.0) Gecko/20100101 Firefox/139.
"
hantale@chantale-VMware-Virtual-Platform:/var/log/apache2$
```

```
0"
chantale@chantale-VMware-Virtual-Platform:/var/log/apache2$ cat site1_error.log
chantale@chantale-VMware-Virtual-Platform:/var/log/apache2$ ls
access.log        error.log.4.gz           site1_access.log.1
access.log.1      error.log.5.gz           site1_access.log.2.gz
access.log.2.gz   error.log.6.gz           site1_error.log
error.log         error.log.7.gz           site1_error.log.1
error.log.1       error.log.8.gz           site2.local_access.log
error.log.10.gz   error.log.9.gz           site2.local_access.log.1
error.log.2.gz    other_vhosts_access.log  site2.local_error.log
error.log.3.gz    site1_access.log         site2.local_error.log.1
chantale@chantale-VMware-Virtual-Platform:/var/log/apache2$
```

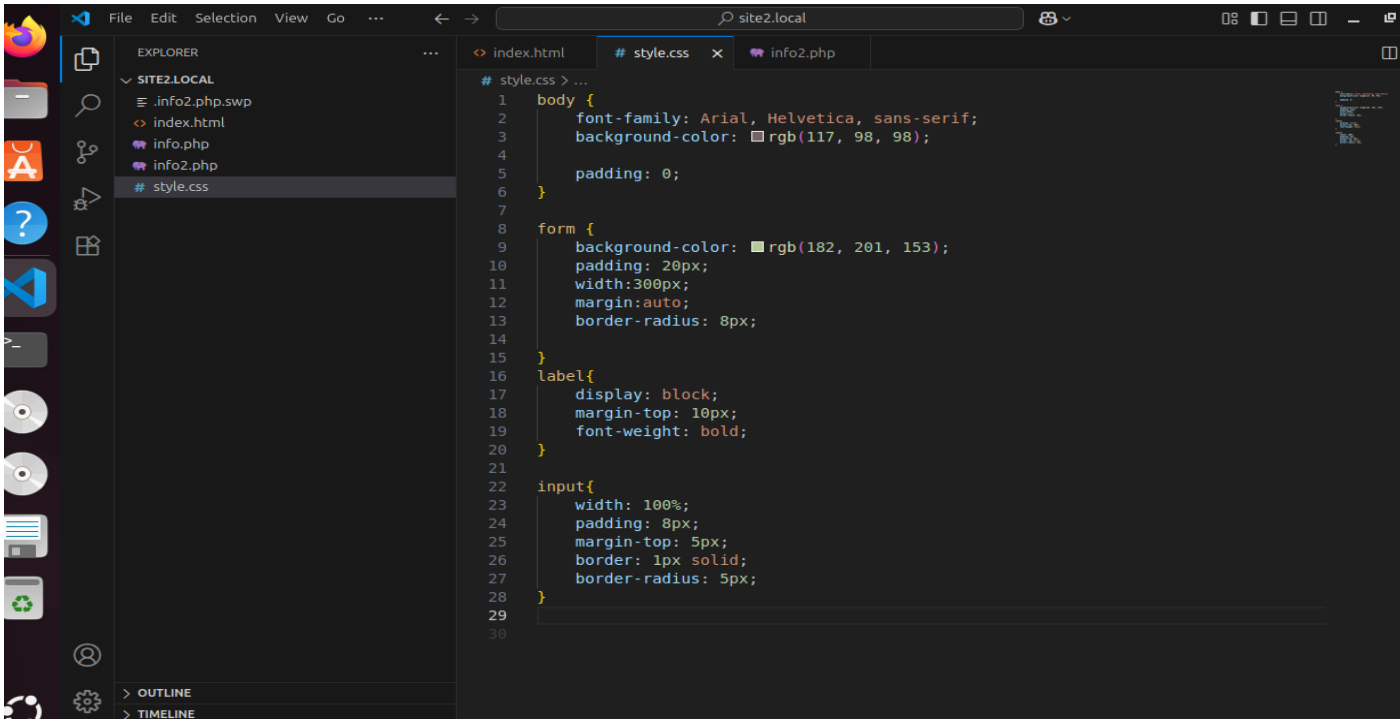
Site2.local

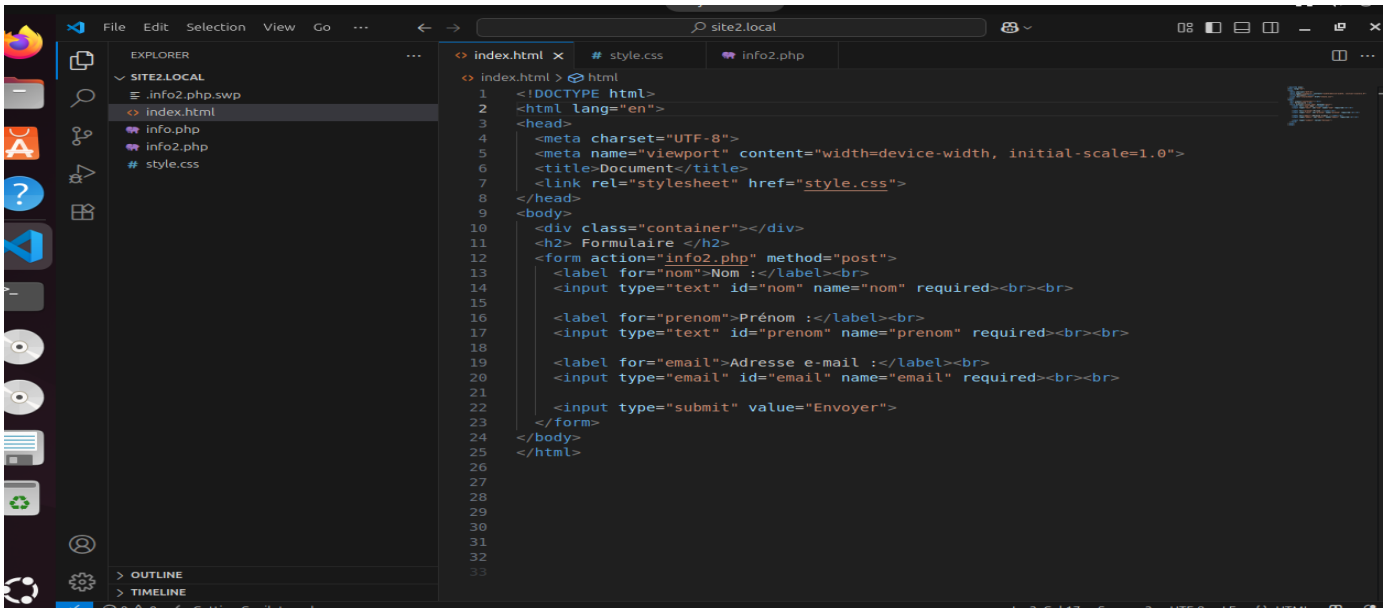
```
chantale@chantale-VMware-Virtual-Platform:/var/log/apache2$ cat site2.local_acc
ss.log
127.0.0.1 - - [15/Jun/2025:11:48:41 +0200] "GET /index.html HTTP/1.1" 200 575 "
" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:139.0) Gecko/20100101 Firefox/139
0"
127.0.0.1 - - [15/Jun/2025:11:48:41 +0200] "GET /favicon.ico HTTP/1.1" 404 489 "
http://site2.local/index.html" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:139.
) Gecko/20100101 Firefox/139.0"
127.0.0.1 - - [15/Jun/2025:11:53:03 +0200] "GET /index.html HTTP/1.1" 200 576 "
" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:139.0) Gecko/20100101 Firefox/139
0"
127.0.0.1 - - [15/Jun/2025:11:53:03 +0200] "GET /index.html HTTP/1.1" 200 575 "
" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:139.0) Gecko/20100101 Firefox/139
0"
127.0.0.1 - - [15/Jun/2025:11:53:04 +0200] "GET /index.html HTTP/1.1" 200 575 "
" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:139.0) Gecko/20100101 Firefox/139
0"
127.0.0.1 - - [15/Jun/2025:11:53:04 +0200] "GET /index.html HTTP/1.1" 200 575 "
" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:139.0) Gecko/20100101 Firefox/139
```

```
antale@chantale-VMware-Virtual-Platform:/var/log/apache2$ cat site2.local_erro
log
un Jun 15 12:12:24.711431 2025] [proxy_fcgi:error] [pid 6111] [client 127.0.0.
56210] AH01071: Got error 'Primary script unknown', referer: http://site2.loca
index.html
un Jun 15 12:12:34.890415 2025] [proxy_fcgi:error] [pid 2174] [client 127.0.0.
42470] AH01071: Got error 'Primary script unknown', referer: http://site2.loca
index.html
un Jun 15 12:12:57.392656 2025] [proxy_fcgi:error] [pid 2172] [client 127.0.0.
35124] AH01071: Got error 'PHP message: PHP Parse error:  syntax error, unexpe
ed variable "$conn" in /var/www/site2.local/info2.php on line 8'
un Jun 15 12:13:01.178773 2025] [proxy_fcgi:error] [pid 6119] [client 127.0.0.
35128] AH01071: Got error 'PHP message: PHP Parse error:  syntax error, unexpe
ed variable "$conn" in /var/www/site2.local/info2.php on line 8'
un Jun 15 13:55:03.641172 2025] [proxy_fcgi:error] [pid 1082] [client 127.0.0.
53684] AH01071: Got error 'PHP message: PHP Parse error:  syntax error, unexpe
ed variable "$conn" in /var/www/site2.local/info2.php on line 8'
un Jun 15 13:55:07.235456 2025] [proxy_fcgi:error] [pid 1078] [client 127.0.0.
42028] AH01071: Got error 'PHP message: PHP Parse error:  syntax error, unexpe
ed variable "$conn" in /var/www/site2.local/info2.php on line 8'
un Jun 15 13:55:09.607219 2025] [proxy_fcgi:error] [pid 1087] [client 127.0.0.
42036] AH01071: Got error 'PHP message: PHP Parse error:  syntax error, unexpe
ed variable "$conn" in /var/www/site2.local/info2.php on line 8'
```

Le client souhaite maintenant avoir un mini formulaire sur le site « «site2.local »

```php
<?php

ini_set('display_errors', 1);
ini_set('display_startup_errors', 1);
error_reporting(E_ALL);


$conn = new mysqli("localhost", "phpmyadmin", "11Decembre@", "phpmyadmin");


if ($conn->connect_error) {
    die("Connexion échouée : " . $conn->connect_error);
}


$nom = $_POST['nom'];
$prenom = $_POST['prenom'];
$email = $_POST['email'];


$sql = "INSERT INTO site2_local (nom, prenom, email) VALUES ('$nom', '$prenom', '$

if ($conn->query($sql) === TRUE) {
    echo "Données enregistrées avec succès !";
} else {
    echo "Erreur : " . $conn->error;
}

$conn->close();
?>
```

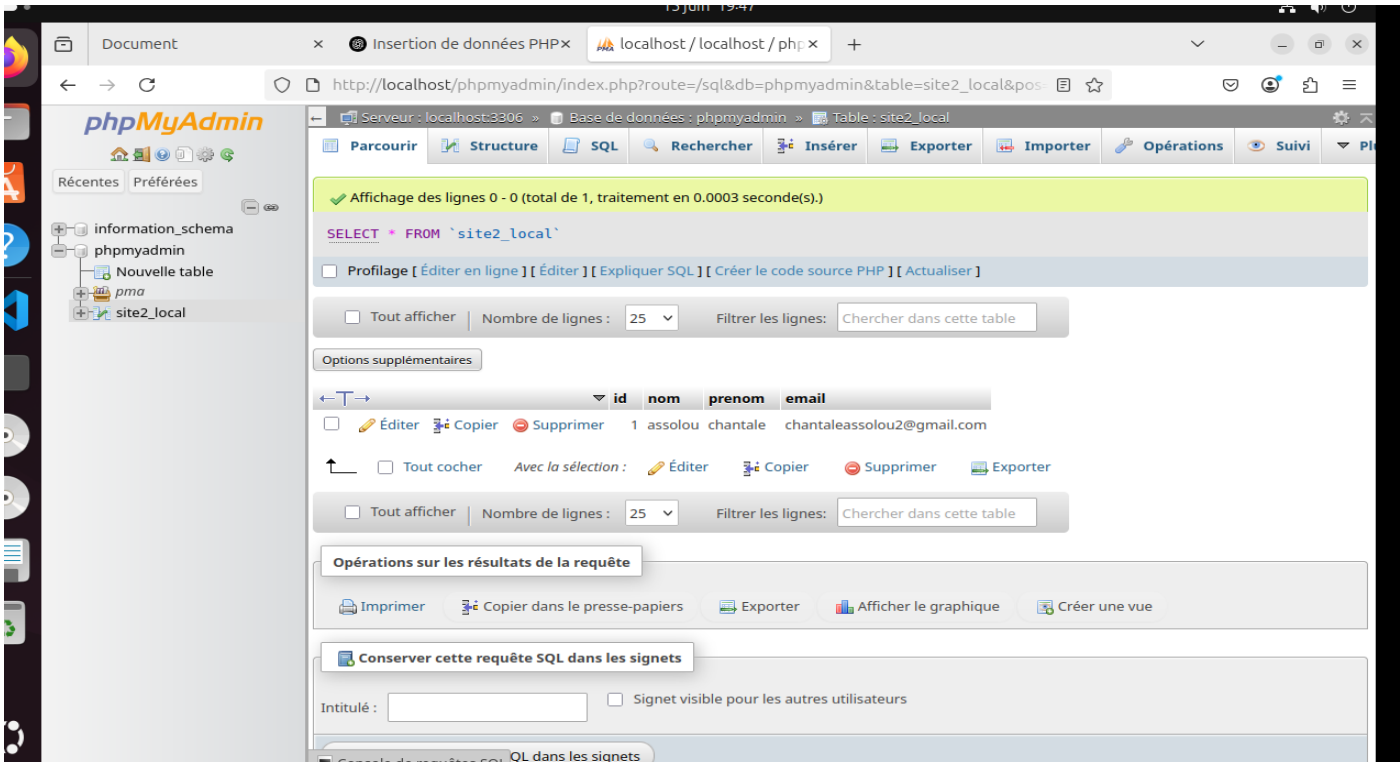

File Edit Selection View Go
site2.local
EXPLORER
SITE2.LOCAL
.info2.php.swp
index.html
info.php
info2.php
style.css
index.html
style.css
info2.php
style.css > ...
body {
font-family: Arial, Helvetica, sans-serif;
background-color: rgb(117, 98, 98);
padding: 0;
}
form {
background-color: rgb(182, 201, 153);
padding: 20px;
width:300px;
margin:auto;
border-radius: 8px;
}
label{
display: block;
margin-top: 10px;
font-weight: bold;
}
input{
width: 100%;
padding: 8px;
margin-top: 5px;
border: 1px solid;
border-radius: 5px;
}
OUTLINE
TIMELINE


File Edit Selection View Go
site2.local
EXPLORER
SITE2.LOCAL
.info2.php.swp
index.html
info.php
info2.php
style.css
index.html
style.css
info2.php
index.html > html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Document</title>
<link rel="stylesheet" href="style.css">
</head>
<body>

<h2> Formulaire </h2>
<form action="info2.php" method="post">
<label for="nom">Nom :</label>
<input type="text" id="nom" name="nom" required>

<label for="prenom">Prénom :</label>
<input type="text" id="prenom" name="prenom" required>

<label for="email">Adresse e-mail :</label>
<input type="email" id="email" name="email" required>

<input type="submit" value="Envoyer">
</form>
</body>
</html>
OUTLINE
TIMELINE

Le formulaire fonctionne bien